

Ex2

1/ Trivial

$$2/ \forall n \in \mathbb{N}, A_n(f) = \frac{2}{\pi} \left(\int_0^{\pi} f(x) \sin(nx) dx \right) \text{ car } x \mapsto f(x) \sin(nx) \text{ est}$$

$$= \frac{2}{\pi} \left(\underbrace{\int_0^{\pi/2} x \sin(nx) dx}_A + \int_{\pi/2}^{\pi} (\pi-x) \sin(nx) dx \right) \text{ parce que } f \text{ et } \sin \text{ sont impaires}$$

$$= \frac{2}{\pi} \left(A + \int_0^{\pi/2} t \sin(n(\pi-t)) dt \right) \text{ si } t = \pi - x$$

$$= \frac{2}{\pi} \left(A + \int_0^{\pi/2} t (\sin n\pi \overset{\text{const}}{=} - \cos n\pi \overset{\text{const}}{=} (-1)^n) dt \right)$$

$$= \frac{2}{\pi} \left(A + \int_0^{\pi/2} t (0 - (-1)^n \sin nt) dt \right)$$

$$\text{Donc } A_n(f) = \frac{2}{\pi} (1 - (-1)^n) \int_0^{\pi/2} x \sin nx dx$$

$$3/ \forall n \in \mathbb{N}, A_{2n}(f) = \frac{2}{\pi} (1-1) \int_0^{\pi/2} x \sin 2nx dx = 0$$

$$4/ \forall n \in \mathbb{N}, A_{2n+1}(f) = \frac{2}{\pi} (1-(-1)^{2n+1}) \int_0^{\pi/2} x \sin(2n+1)x dx$$

$$3/ \forall n \in \mathbb{N}, A_{2n}(f) = \frac{2}{\pi} (1-1) \int_0^{\pi/2} x \sin(2nx) dx = 0$$

$$\begin{aligned} 4/ \forall n \in \mathbb{N}, A_{2n+1}(f) &= \frac{2}{\pi} \frac{(1+1)}{(2n+1)} \int_0^{\pi/2} x \cos((2n+1)x) dx \\ &= \frac{-4}{\pi(2n+1)} \left(\int_0^{\pi/2} x \cos((2n+1)x) dx - \int_0^{\pi/2} \cos((2n+1)x) dx \right) \\ &= \frac{-4}{\pi(2n+1)} \left(0 - 0 - \frac{1}{(2n+1)} \int_0^{\pi/2} \cos((2n+1)x) dx \right) \\ A_{2n+1}(f) &= \frac{4}{\pi(2n+1)^2} (1n^2 - 0) \end{aligned}$$

$$5/a) \forall x \in \mathbb{R}, f(x) = \sum_{k=0}^{\infty} A_{2k}(f) \sin(2kx) + \sum_{k=0}^{\infty} A_{2k+1}(f) \sin((2k+1)x)$$

$$f(x) = \sum_{k=0}^{\infty} \frac{4(1k)^2}{\pi(2k+1)^2} \sin((2k+1)x)$$

On veut de se débarrasser de $(1k)^2$ on va prendre $x = \pi/4$

$$\text{Dans ce cas } f(\pi/4) = \frac{\pi}{4} = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(1k)^2}{(2k+1)^2} \text{ d'où } \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$$

$$b) \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} + \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \frac{\pi^2}{8} + \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2} \text{ d'où } \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{8} \cdot \frac{4}{3} = \frac{\pi^2}{6}$$

$$\sum_{n=1}^{\infty} \frac{(1n)^{2n-1}}{n^2} = \sum_{k=0}^{\infty} \frac{(1k)^{2k+1-1}}{(2k+1)^2} + \sum_{k=1}^{\infty} \frac{(1k)^{2k-1}}{(2k)^2} = \frac{\pi^2}{8} + \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{8} + \frac{1}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12}$$